「Sign of permutation.

Let a polynomial over $\{\pm 1\}$ with distinct n-variables $x_1, \cdots, x_n$ as

$$\Delta \overset{\text{def}}{=} \prod_{1 \le i < j \le n} (x_i - x_j)$$

For each $\sigma \in S_n$, let σ act on Δ by permuting:

$$\sigma(\Delta) = \prod_{1 \le i < j \le n} (x_{\sigma(i)} - x_{\sigma(j)})$$

Since σ is bijective, there are two possibities ;

$$\sigma(\Delta) = \pm \Delta.$$

Now, let

$$\text{sign} : S_n \to \{\pm 1\} : \sigma \mapsto \begin{cases} 1 & \text{if } \sigma(\Delta) = \Delta \\ -1 & \text{if } \sigma(\Delta) = -\Delta. \end{cases}$$

「Prop. Sign is a Surjective Homomorphism.

Proof. Given $\sigma, \tau \in S_n$,

$$(\tau \circ \sigma)(\Delta) = \prod_{1 \le i < j \le n} (x_{(\tau\circ\sigma)(i)} - x_{(\tau\circ\sigma)(j)})$$

Suppose $\sigma(\Delta)$ has exactly k factors of $(x_j - x_i)$ with $j > i$. That is $\text{sign}\,\sigma = (-1)^k$.

So,

$$\sigma(\Delta) = \prod_{1 \le i < j \le n} (x_{\sigma(i)} - x_{\sigma(j)}) = \text{sign}\,\sigma \cdot \prod_{1 \le p < q \le n} (x_p - x_q)$$

Therefore, $(\tau \circ \sigma)(\Delta) = \text{sign}\,\sigma \cdot \prod_{1 \le p < q \le n} (x_{\tau(p)} - x_{\tau(q)})$, it follows

$$\text{sign}(\tau \circ \sigma) = \text{sign}\,\sigma \cdot \text{sign}\,\tau.$$

Surjectiveness; Clearly, $\text{sign}(id) = 1$. And, $\text{sign}((1,2)) = -1$, So clear. ∎

「Prop. $\ker \text{sign} = A_n$.

「Prop. m-cycle is odd iff m is even

「Prop. a permutation σ is odd iff number of cycles of even length is odd, in decomposition.

Prop. S_n is generated by $\{(i \ \ i+1) \mid 1 \le i \le n-1\}$.

Proof. Given $j \in \{2, \cdots, n-1\}$,

$$(1 \ 2 \ 3 \cdots n) = (1 \ n)(1 \ n-1)\cdots(1 \ 3)(1 \ 2)$$

$$(1 \ 2)(2 \ 3)\cdots(j-1 \ \ j) = (1 \ 2 \ 3 \cdots j)$$

So, the generated set contains $(1 \ 2 \cdots j)$ for all $2 \le j \le n$.

Moreover, observe that

$$(2 \ 3)(1 \ 2)(2 \ 3)^{-1} = (2 \ 3)(1 \ 2 \ 3) = (1 \ 3)$$

$$(3 \ 4)(1 \ 3)(3 \ 4) = (3 \ 4)(1 \ 3 \ 4) = (1 \ 4)$$

$$\vdots$$

$$(j \ \ j+1)(1 \ j)(j \ \ j+1) = (1 \ \ j)$$

Now, for any $1 \le j < k \le n$,

$$(j \ \ k) = (1 \ k)(1 \ j)$$

can be generated by the given set. All permutations can be represented by product of transpositions, so $S_n = \langle (i \ \ i+1) \mid 1 \le i \le n-1 \rangle$

Corollary. $S_n = \langle (1 \ 2), (1 \ 2 \cdots n) \rangle$

Proof. observe that

$$(1 \ 2 \cdots n)(1 \ 2)(1 \ 2 \cdots n)^{-1} = (1 \ 2 \cdots n)(1 \ 2)(n \ \ n-1 \cdots 1) = (2 \ 3)$$

$$(1 \ 2 \cdots n)(2 \ 3)(1 \ 2 \cdots n)^{-1} = (3 \ 4)$$

$$\vdots$$

So, $\langle (1, 2), (1 \ 2 \cdots n) \rangle$ contains $(i \ \ i+1)$ for all $1 \le i \le n-1$, proved

+ Note that

$$\begin{aligned}
&(a_1 \ a_2 \ \cdots \ a_n) \\
&= (a_1 \ a_n)(a_1 \ a_2 \ \cdots \ a_{n-1}) \\
&= (a_1 \ a_n)(a_1 \ a_{n-1})(a_1 \ a_2 \cdots a_{n-2}) \\
&\ \ \vdots \\
&= (a_1 \ a_n)(a_1 \ a_{n-1}) \cdots (a_1 \ a_2)
\end{aligned}$$

Prop. For a prime number p, S_p is generated by any one transposition and p-cycle.

proof. WLOG, let the one transposition is $(1\ x)$ and p-cycle is $(1 \cdots p)$.
where $x \in \{2, \cdots, p\}$. And, since p is prime number,

$$(1\ 2 \cdots p)^k \quad (1 \le k \le p-1)$$

are all p-cycles. And, take $k = x-1$, then

$$(1\ 2 \cdots p)^k = (1\ x \cdots)$$

So $\langle (1\ x), (1 \cdots p) \rangle = \langle (1\ x), (1\ x \cdots) \rangle = S_p$.

Note; $\langle (1\ 3), (1\ 2\ 3\ 4) \rangle$ cannot generate S_4. observe that

$(1\ 3)^2 = id$, $(1\ 2\ 3\ 4)^2 = (1\ 3)(2\ 4)$, $(1\ 2\ 3\ 4)^3 = (1\ 4\ 3\ 2)$

$(1\ 3)(1\ 2\ 3\ 4) = (1\ 2)(3\ 4)$

$(1\ 3)(1\ 2\ 3\ 4)^2 = (2\ 4)$

$(1\ 3)(1\ 2\ 3\ 4)^3 = (1\ 4)(2\ 3)$

So, $\langle (1\ 3), (1\ 2\ 3\ 4) \rangle = \langle (1\ 2)(3\ 4), (1\ 3)(2\ 4) \rangle$.

Prop. A_4 has No subgroup of order 6.

Prof. Let $V = \{id, (12)(34), (13)(24), (14)(23)\}$ be a subset of A_4.
Except $id \in V$, the elements of V have order of 2, exactly,
these are all elements of order 2 in A_4.
The Conjugate cannot change the order and parity, V is a Normal subgroup of A_4
And, since A_4/V has order of 3, it must be contain order 3 element,
so, it must be isomorphic to $\mathbb{Z}_3$, cyclic.
Meanwhile, if there exists a order of 6 subgroup, N, then
since $[A_4 : N] = 2$, thus for any $g \in A_4$, $gN = Ng$. so, it is normal.
Now, $A_4/N \cong \mathbb{Z}_2$, cyclic.
Meanwhile, observe that; for $G' = \{x^{-1}y^{-1}xy \mid x, y \in G\}$, and $N \trianglelefteq G$,

$$G/N \text{ is abelian iff } G' \leq N$$

To show this, Let $x^{-1}y^{-1}xy \in G'$ be given. Since G/N is abelian,

$$x^{-1}y^{-1}xyN = x^{-1}N\,y^{-1}N\,xN\,yN = x^{-1}N\,xN\,y^{-1}N\,yN = N$$

so $x^{-1}y^{-1}xy \in N$.

Therefore, $G' \trianglelefteq N$ and $G' \trianglelefteq V$. And, since A_4 is Not abelian, $G' \neq 1$.
And, if $|G'| = 2$, then it must be Contain an order of 2 element,
but since G' is normal, then it has all order of 2 element.
Thus, $G' \supseteq V$, or $G' = V$. Now, $|G'| = 4$ but $G' < N$,
it is Contradiction to the Lagrange Thm.